Sam Voncken

Automotive Engineer — Motorsport Simulation & Performance

Graduated automotive engineer, now specialising in simulation & analysis for motorsport at Lund University. Focus: thermal modelling, combustion engines and hybrid power units, with MATLAB/Simulink, CFD and CAD as daily tools — backed by hands-on engine-dyno testing and measurement, 1D engine simulation, and models calibrated and validated against experimental data. Currently cooling system engineer at Lund Formula Student (powertrain group) — ranked top 37 worldwide. **Goal: a master's thesis in motorsport performance engineering — power unit or thermal simulation & analysis.**

EXPERIENCE

Cooling System Engineer — Dec 2025 – present

Lund Formula Student — Lund, Sweden

- Developing a thermal model of the LFS27 accumulator (80 kW full-electric formula car, 4 hub motors): air-cooling sized for a 550 W heat load; hand calculations correlated against Ansys thermal simulations.
- Designed and manufactured the cooling-loop swirl pot (de-aeration) for LFS26, maintaining swirl under cornering and braking G-forces at 23 L/min.
- Accumulator-cooling sub-team within Powertrain; team of ~120 members, ranked top 37 worldwide (2 positions higher in 2026).

Research Engineer (extra-curricular) — Oct 2025 – present

Project LoCaGas, LTH — locagas.eu

- EU-funded project (Interreg South Baltic): experimental research on a single-cylinder-converted Scania D13 heavy-duty engine running on landfill gas with an RME–pyrolysis-oil pilot blend.
- Plan and execute engine-dyno test campaigns — from measurement planning to post-processing — optimising combustion efficiency of the diesel-to-landfill-gas conversion.

Combustion Knock Analysis — Sep 2023 – Jun 2024

Westport Fuel Systems — BEng graduation thesis

- Investigated knock detection on heavy-duty dual-fuel engines, comparing structure-borne knock-sensor signals against an in-cylinder piezo reference (DEWETRON / DEWESoft).
- Built MATLAB post-processing and data-visualization scripts; calibrated and validated knock-threshold conditioning against experimental data (Plex Knock Monitor V3).
- Determined the optimal knock-sensor position (block–head mating surface) — early-knock detection on par with the in-cylinder reference.

Durability Test Engineer — Internship — Sep 2022 – Mar 2023

Lightyear — Helmond, Netherlands

- Investigated a damage concern on the 3-phase cable of the Lightyear Zero hub motor (required lifetime 10 years / 250,000 km); defined, built and programmed (WINCAPS III) an accelerated durability rig around a DENSO robot arm — full cycle under 10 s, targeting 500,000 cycles in one week.
- The cable did not meet vehicle lifetime — a safety-critical finding with service-interval recommendations.

Engine Data Analysis — Fuel Efficiency — Sep 2020 – Feb 2021

RAVO street sweepers — Minor Internal Combustion Engines, HAN

- Developed a MATLAB algorithm to extract fuel-consumption (BSFC) maps from raw logged ECU data; validated the maps against measurements.
- Proposed engine operating points to maximise fuel efficiency — data-driven performance optimisation of a production diesel engine.

CONTACT

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SIMULATION & DATA

CAD & FEA — SolidWorks, Abaqus, Ansys, Creo Parametric

CFD — OpenFOAM, Ansys

Data & scripting — MATLAB & Simulink, Python, Power BI (basics)

Engine simulation — GT-Power, Cantera 1D, AVL BOOST

Engine management — MaxxECU MTune

Measurement — DEWESoft (DEWETRON combustion analyser), Plex Knock Analyser

Robotics — WINCAPS III (DENSO robot arms)

Web — HTML/CSS basics, GitHub, Cloudflare

PROJECTS & VENTURES

Founder Retroneering.com — Jul 2026 – present

Retroneering.com

- Founded a company reverse-engineering discontinued parts for classic and race cars — CAD, GD&T drawings, FEA validation.
- Precision hand-measurement based; in-house photogrammetry (photo-to-3D) pipeline in development. First customers lined up.

Part & Tool Design Engineer — Aug 2024 – Jul 2025

R-Quatre.nl — classic Renault parts

- Designed 24 replacement parts and 2 specialty tools for classic Renault (folded sheet metal, cast selector forks, rubber-moulded, 3D-printed) — from design to manufacturer liaison and webshop sales.
- First production batch of 200 tools nearly sold out within 12 months; parts shipped worldwide, from across Europe to Madagascar and Australia.

Founder, Team Manager & Engineer — Apr 2019 – Jun 2022

HAN 4L — student racing team

- Founded and led a racing team of 11 members, 10 nationalities, 3 programmes; still competing today, now 6th generation with 3 cars.
- Rebuilt a Renault 4 for the 4L Trophy humanitarian rally: ~8,000 km Netherlands–Morocco, top-10 international teams finish.
- Raised €15,000 in sponsorship (incl. DENSO); delivered 8 kg of school supplies and food to Sahara communities.
- Designed and flow-simulated an equal-length exhaust header to improve the engine's thermodynamic efficiency.

Personal Rally Car Build — May 2021 – present

Renault R4 Alpine Turbo

- Rebuilt and upgraded a Renault R4 into a rally car: turbocharged powertrain, 5-speed gearbox, 34 hp → 170 hp.
- Programmed a standalone MaxxECU engine-management system — injection and ignition control from multiple sensor inputs.
- Designed and manufactured a CAD/FEA-validated chromoly roll cage (FIA Appendix J Art. 283-8) — current roll cage is 28 kg vs 46 kg previous roll cage at HAN4L (-39%).
- Long-term goals: hybrid powertrain and the Dakar Classic (next step: hub-dyno calibration and performance testing).

Structural Design Engineer — Feb 2021 – Jun 2021

EBUSCO 3.0 electric bus — HAN specialisation project

- Designed a thermoplastic-composite alternative to the cast-iron axle bridge of an electric bus (biomimicry, topology optimisation) — ~40% weight reduction.

EDUCATION

MSc Sustainable Energy Engineering — 2025 – 2027

Lund University (LTH) — Lund, Sweden

- Courses: Advanced Combustion Engines, Hybrid Vehicle Technology, Hydrogen & Fuel Cells, CFD & Heat Transfer, Aerodynamics.
- Master's thesis (30 ECTS) at a motorsport company, from January 2027.

BEng Automotive Engineering — 2018 – 2024

HAN University of Applied Sciences — Arnhem, Netherlands

- Specialisation in Structural Design, Manufacturing and Powertrain; extra-curricular minors in Internal Combustion Engines and Lightweight Vehicle Design.

Project portfolio (roll cage, R-Quatre parts, simulations, Formula Student) on request.

MANUFACTURING

Welding · Milling · Lathing · Laser & waterjet cutting · 3D printing & scanning · Aluminium casting · Fiberglass composite moulding · Vacuum thermoforming · Classic car restoration & upgrading

LANGUAGES

Dutch — native

English — fluent

Italian — learning

CERTIFICATIONS

Garrett Turbos — Racing & Performance Training (2020)

CSWA Mechanical Design (2019)

Introduction to Abaqus & FEA (2022)

Safety for Operational Supervisors (2021)

First Aid & Resuscitation (2022)

AWARDS

KNAW Education Award (Netherlands) · ICYS (Serbia) · OKSEEF (Turkey)

— award-winning high-school engineering thesis, 2018